

| National University of Computer and Emerging Sciences, Lahore Campus              |             |                                   |              |           |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------|--------------|-----------|
|  | Course:     | Analog and Digital Communications | Course Code: | EE323     |
|                                                                                   | Program:    | BS(Electrical Engineering)        | Semester:    | Fall 2018 |
|                                                                                   | Duration:   | 1 Hour                            | Total Marks: | 45        |
|                                                                                   | Paper Date: | 16-Nov-18                         | Weight       | 20%       |
|                                                                                   | Section:    | ALL                               | Page(s):     | 2         |
| Exam:                                                                             |             | MID-II                            | Roll No:     |           |
| Instruction/Notes: Attempt all parts of a question together.                      |             |                                   |              |           |

**Question # 1 [CLO: 4]**

[15]

A vestigial shaping filter  $H_i(\omega)$  has a transfer function as shown in figure 1. The carrier frequency is  $f_c = 100\text{kHz}$  and the baseband signal bandwidth is  $5\text{kHz}$ .

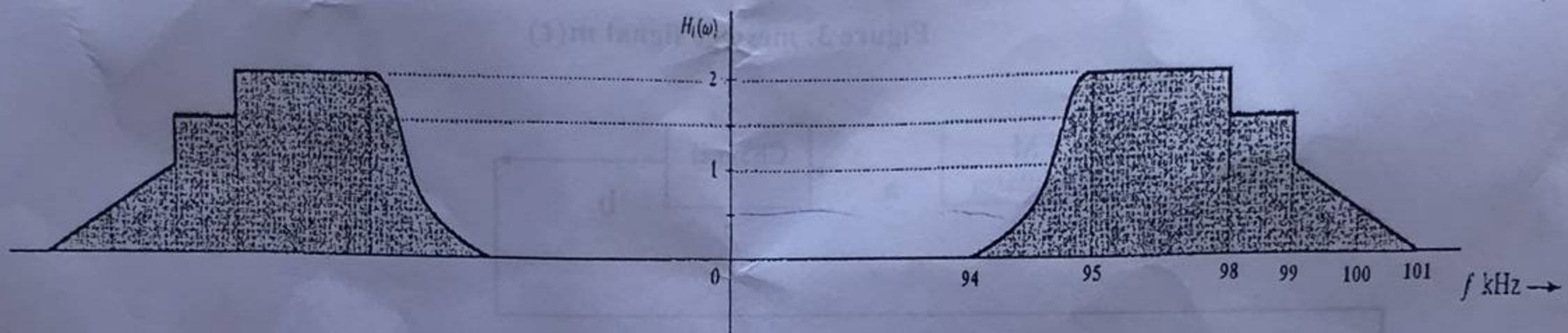


Figure 1: Vestigial shaping filter

- Find the bandwidth of the VSB signal.
- Sketch the necessary equalizer filter response  $H_o(\omega)$  for distortion-less reception.
- Design and illustrate a receiver system block diagram for VSB.

**Question # 2 [CLO: 5]**

[15]

A signal  $m(t) = 2 \cos(2\pi * 10^3 t)$  phase modulates (PM) a  $1\text{MHz}$ ,  $2V_{P-P}$  carrier to produce a frequency deviation of  $4\text{kHz}$ .

- Write the time-domain expression for the resulting PM signal  $\phi_{PM}(t)$ ?
- What is the value of phase constant  $k_p$ ?
- What is phase deviation  $\Delta\phi$ ?
- What is the bandwidth of the PM signal?
- Can we demodulate  $\phi_{PM}(t)$  using demodulator shown in figure 2, if yes/no explain your answer with reasoning. Suggest modification for given demodulator if required? Also write equation for  $e_o(t)$ .

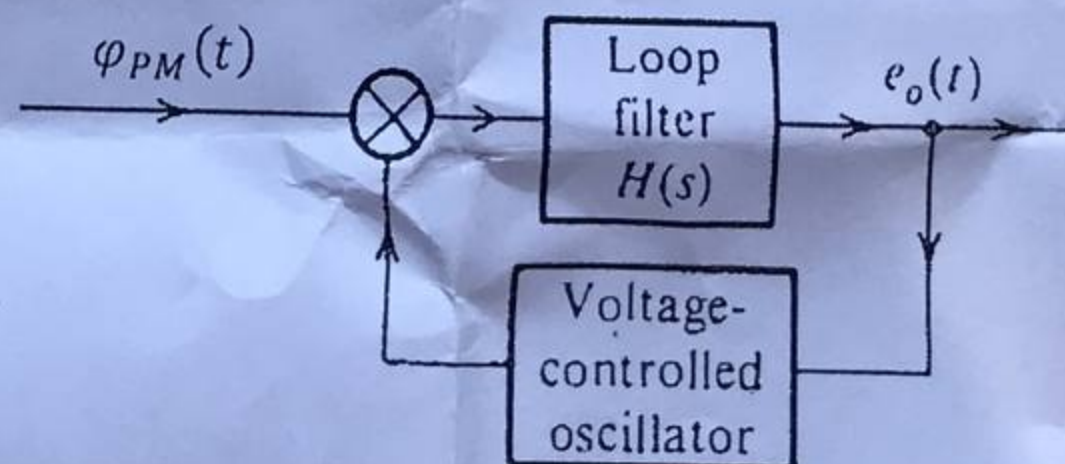


Figure 2: Demodulaor

**Question # 3 [CLO: 5]**

[15]

A message signal  $m(t)$ , shown in figure 3, frequency-modulates a carrier of frequency  $f_c = 10\text{MHz}$  with  $k_f = 2\pi * 10^5\text{Hz}$ . The carrier amplitude is  $A$ . When the modulated signal is passed through the channel its amplitude is changed randomly. The received signal is demodulated by passing through slope detector as shown in figure 4.

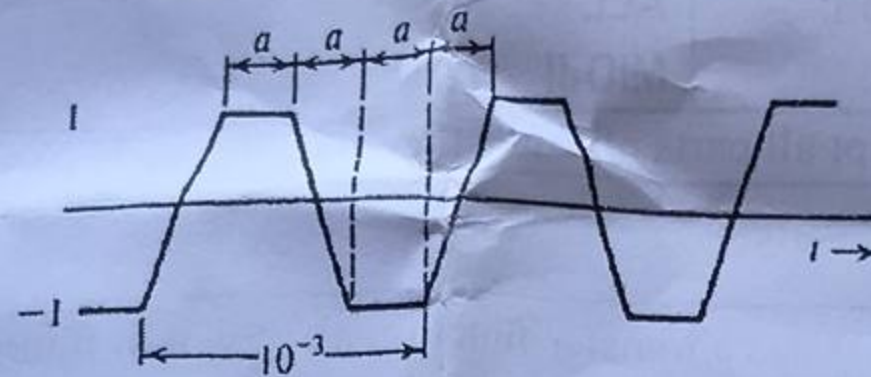


Figure 3: message signal  $m(t)$

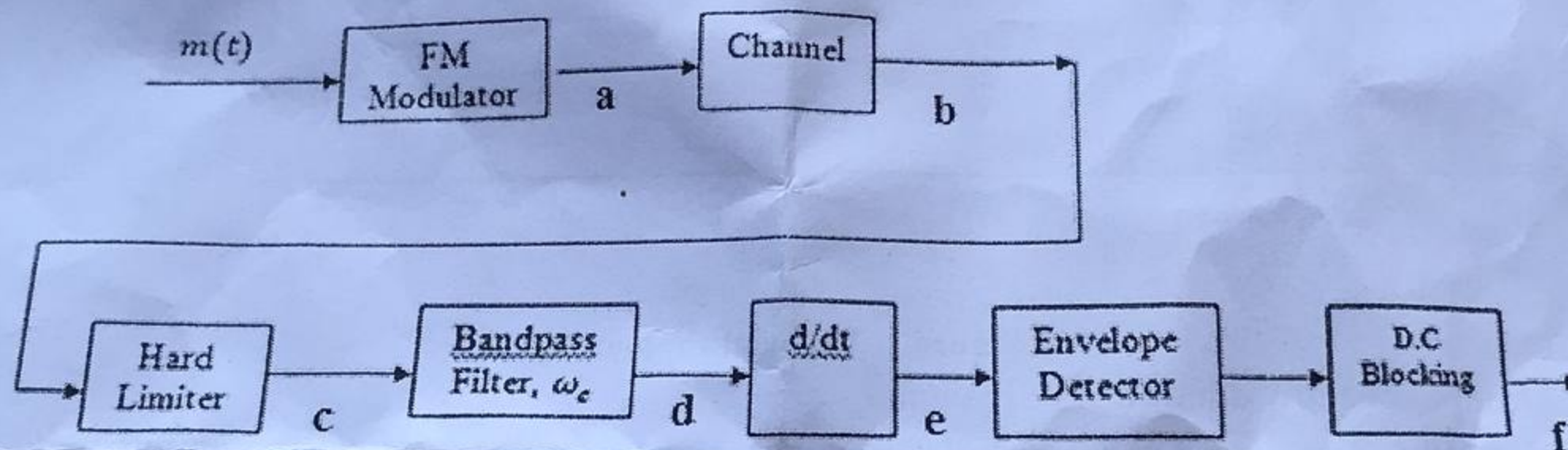


Figure 4: FM modulator and demodulator

- Sketch the waveform, approximated, at points a, b, c, d, e and f.
- What are the values of  $f_c$  and  $\Delta f$  at point a?
- Estimate the bandwidth of the FM signal. Assume the bandwidth of  $m(t)$  to be the third harmonic frequency of  $m(t)$ .